

Finite-Dimensional Quotients Obstruct Epsilon-Hypercyclicity

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Status

This packet records a candidate partial result, likely valid. It does not construct a new ε -hypercyclic non-hypercyclic operator. Instead it isolates a general obstruction which any solution to the open problem in Tapia-Garcia, arXiv:2103.08075, must avoid.

Source Context

Tapia-Garcia studies ε -hypercyclic operators. For a Banach space X , an operator $T \in \mathcal{L}(X)$ is ε -hypercyclic if there is a vector $x \in X$ such that, for every nonzero $y \in X$, some $n \geq 0$ satisfies

$$\|T^n x - y\| \leq \varepsilon \|y\|.$$

The introduction asks whether every separable infinite-dimensional Banach space admits an ε -hypercyclic operator which is not hypercyclic, with $\varepsilon \in (0, 1)$.

$n \in \mathbb{N}$ for which

$$\|T^n x - y\| \leq \varepsilon \|y\|.$$

The vector x is said to be an ε -hypercyclic vector of T .

By the very definition, it is clear that every hypercyclic operator is ε -hypercyclic for every $\varepsilon > 0$. So, we already know that ε -hypercyclic operators exist in every separable infinite dimensional Banach space, see [2] and [8]. Also, every linear operator is 1-hypercyclic, since 0 is a 1-hypercyclic vector. However, up to the best of our knowledge, it remains open the question if every separable infinite dimensional Banach space admits an ε -hypercyclic which is not hypercyclic (with $\varepsilon \in (0, 1)$). Up to the best of our knowledge, in the literature we can find the construction of such an operator in the spaces $\ell^1(\mathbb{N})$ and $\ell^2(\mathbb{N})$, see [3] and [4] respectively. We point out that the technique used by Bayart in [4] is motivated by the one used by Badea, Grivaux and Müller in [3]. Also, we can find other mentions of ε -hypercyclicity in [5, 7, 18].

In this work, we introduce the following ε -Hypercyclicity criterion (Theorem 1.2) and we use it to provide several examples of ε -hypercyclic operators which are not hypercyclic

The source paper gives positive constructions in several classical cases, especially when the space contains a complemented subspace of the form needed for its operator-weighted shift construction. The result below records a different, negative constraint on possible constructions.

Result

Lemma 1 (No finite-dimensional ε -hypercyclic operators). *Let $F \neq \{0\}$ be a finite-dimensional real or complex Banach space. If $A \in \mathcal{L}(F)$ and $0 \leq \varepsilon < 1$, then A is not ε -hypercyclic.*

Proof. It is enough to argue over the complexification in the real case. Suppose that some $u \in F$ were an ε -hypercyclic vector for A . Fix any nonzero $v \in F$. Applying the definition to the targets tv , with $t > 0$, gives integers $n(t)$ such that

$$\|A^{n(t)}u - tv\| \leq \varepsilon t\|v\|.$$

Thus

$$(1 - \varepsilon)t\|v\| \leq \|A^{n(t)}u\| \leq (1 + \varepsilon)t\|v\|.$$

Letting $t \downarrow 0$ and $t \uparrow \infty$, the orbit $(A^n u)$ must have a subsequence converging to 0 and must also be unbounded.

Let $E = \text{span}\{A^n u : n \geq 0\}$, a finite-dimensional A -invariant cyclic subspace. Since $A^{n_j}u \rightarrow 0$ along a subsequence, the restriction $A_E = A|_E$ cannot have an eigenvalue λ with $|\lambda| \geq 1$. Indeed, if $A_E^* \varphi = \lambda \varphi$ for some nonzero $\varphi \in E^*$, then cyclicity of u in E gives $\varphi(u) \neq 0$; otherwise φ would vanish on every $A^n u$ and hence on all of E . But then

$$|\varphi(A^n u)| = |\lambda|^n |\varphi(u)|$$

cannot tend to 0 along any subsequence. Therefore all eigenvalues of A_E have modulus strictly less than 1, so $A_E^n u \rightarrow 0$. In particular the orbit is bounded, contradicting the unboundedness forced above. $\square$

Proposition 1 (Epsilon-hypercyclicity passes to quotients with loss). *Let $q : X \rightarrow Y$ be a quotient map between Banach spaces, and suppose $T \in \mathcal{L}(X)$, $S \in \mathcal{L}(Y)$ satisfy $qT = Sq$. If x is an ε -hypercyclic vector for T , then qx is a δ -hypercyclic vector for S for every $\delta > \varepsilon$.*

Proof. Fix $\delta > \varepsilon$ and $0 \neq z \in Y$. If $\varepsilon = 0$, choose any preimage $y \in X$ of z . Otherwise choose $\eta > 0$ such that $\varepsilon(1 + \eta) < \delta$, and then choose $y \in X$ with $qy = z$ and $\|y\| \leq (1 + \eta)\|z\|$, using the quotient norm. By ε -hypercyclicity there is n such that

$$\|T^n x - y\| \leq \varepsilon\|y\|.$$

Applying q gives

$$\|S^n qx - z\| = \|q(T^n x - y)\| \leq \varepsilon\|y\| < \delta\|z\|$$

in the case $\varepsilon > 0$, and the same conclusion with error 0 in the case $\varepsilon = 0$. $\square$

Corollary 1 (Finite-dimensional quotient obstruction). *Let X be a Banach space, $T \in \mathcal{L}(X)$, and $0 \leq \varepsilon < 1$. If T is ε -hypercyclic, then there is no closed T -invariant subspace $M \subset X$ for which X/M is nonzero and finite-dimensional. Equivalently, T has no nonzero finite-dimensional invariant quotient.*

Proof. If such an M existed, T would induce an operator $\bar{T}$ on the nonzero finite-dimensional quotient X/M . Choose δ with $\varepsilon < \delta < 1$. Proposition 1 would make $\bar{T}$ δ -hypercyclic, contradicting Lemma 1. $\square$

Proof Intuition

The point is that $\varepsilon < 1$ is strong enough to see scale. If an orbit approximates every nonzero target within relative error ε , then along the fixed ray $\{tv : t > 0\}$ its norms must come arbitrarily close to zero and also become arbitrarily large. Finite-dimensional linear dynamics cannot do both. A

subsequence tending to zero rules out every spectral component of modulus at least one on the cyclic subspace generated by the vector; hence the orbit actually tends to zero and is bounded.

The quotient statement then says that any ε -hypercyclic construction with $\varepsilon < 1$ must remain δ -hypercyclic on every induced quotient for all $\delta > \varepsilon$. Since finite-dimensional quotients cannot have this property, finite-dimensional marker coordinates cannot be used to force non-hypercyclicity.

Implication for the Source Problem

This does not answer Tapia-Garcia's question. It does, however, remove a natural route: decomposing a Banach space as a hyperplane plus a scalar coordinate, making the hyperplane part hypercyclic, and using the scalar part as a non-hypercyclic obstruction. Such an operator would have a nonzero finite-dimensional invariant quotient, so it cannot be ε -hypercyclic for any $\varepsilon < 1$.

Thus any full construction on arbitrary separable Banach spaces must either build the non-hypercyclic obstruction in an infinite-dimensional quotient or subspace, or use a construction whose failure of hypercyclicity is not detected by a finite-dimensional quotient.

Verification and Novelty Check

The proof is non-computational. The local run indexes were searched for the arXiv id, epsilon-hypercyclicity terms, and finite-dimensional quotient terms; no duplicate packet or attempt for this obstruction was found. Web searches for the same exact themes found the source paper and Bayart's later sharp-threshold construction, but no exact occurrence of this finite-dimensional quotient obstruction.

The novelty confidence is therefore moderate rather than high: the observation is elementary and may be known folklore in linear dynamics, but it was not visible in the bounded search and is directly useful for steering further work on the source problem.

References

- [1] S. Tapia-Garcia, *An epsilon-hypercyclicity criterion and its application on classical Banach spaces*, arXiv:2103.08075, 2021.
- [2] F. Bayart, *ε -hypercyclic operators that are not δ -hypercyclic for $\delta < \varepsilon$* , arXiv:2303.16677, 2023.